# Pair-Preserving Stability in Markets with Entry (and Exit)

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**Abstract**

This paper studies the stability of matchings in two-sided markets subject to uncertain entry (and exit). We introduce pair-preserving stability, which requires that a stable matching admit a stable extension in the perturbed market that preserves all originally matched pairs whose members remain. When only men enter, we show that the woman-optimal stable matching is maximally pair-preserving stable: if it admits no stable extension that preserves its matched pairs, then no other stable matching is pair-preserving stable. This dominance is uniform over all possible realizations of the number or preferences of new entrants and is independent of the size or preference profile of the initial market. We further characterize existence by showing that a pair-preserving stable matching exists if and only if a single critical matching is stable in the perturbed market. We then consider entry on both sides of the market. When entry is asymmetric, with one side entry dominating the other, we characterize the stable matching that maximizes the probability of being pair-preserving stable. Together, these results inform the choice of stable matchings in markets subject to uncertain growth.

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# 1 Introduction

Centralized matching has been a cornerstone of matching theory and has enjoyed many applications, including the design of labor, housing, and education markets.

This paper studies a two-sided one-to-one matching market. Formally, a two-sided one-to-one *market* can be described by $(M, W, p)$, where $M$ and $W$ are groups of agents and $p$ is the collection of agents' preferences.

Following Gale and Shapley [1962], a matching is stable if it is individually rational and admits no blocking pair.

Standard stability is defined with respect to a fixed market. In many applications, however, markets evolve over time through the entry or exit of agents, and a planner may face uncertainty about future market realizations. This paper studies how to select a stable matching today when the market may expand tomorrow.

We introduce pair-preserving stability. A stable matching is pair-preserving stable if it admits a stable extension in the perturbed market that preserves all originally matched pairs whose members remain. This notion captures robustness of stability with respect to market growth when previously formed matches cannot be revised. Such irreversibility may arise from binding contracts, or more generally from the frictions and coordination costs associated with disrupting existing matches.

When there is only men entry, we show that the woman-optimal stable matching is maximally pair-preserving stable: if it admits no stable extension that preserves its matched pairs in the perturbed market, then no other stable matching is pair-preserving stable. We further show that a pair-preserving stable matching exists if and only if a single critical extension is stable, yielding a tractable criterion for planners.

We then consider entry on both sides of the market. In this case, no stable matching is uniformly most pair-preserving. Under asymmetric entry, when one side's entry dominates the other's, we characterize the stable matching that maximizes the probability of admitting a stable pair-preserving extension.

This paper contributes to the literature on robustness of stable matchings. Existing work studies robustness of stable matchings to uncertainties in agents' preferences (Aziz et al. [2020], Menon and Larson [2018]), errors in reported preferences (Mai and Vazirani [2020], Boehmer and Boehmer [2024]), and exits of agents or other disruptions on matched pairs (Boehmer and Boehmer [2024], Jacobovic [2018], Genc et al.

[2017], Genc et al. [2019]). In contrast, we focus on robustness of stability to agent entry while requiring preservation of existing matched pairs. Our approach differs from dynamic matching models with irreversible matchings (Doval [2022]) by maintaining a static notion of stability evaluated after uncertain market expansion.

The remainder of the paper is organized as follows. Section 2 presents the model and the definitions. Section 3 studies one-sided entry (and exit subject to some constraints). Section 4 considers entry on both sides of the market.

# 2 The Model

We consider a one-to-one matching market with two finite sides $M$ and $W$, where $M = \{m_1, \ldots, m_n\}$ and $W = \{w_1, \ldots, w_k\}$. Each agent $i \in M \cup W$ has a complete and strict preference order $\succ_i$ over the opposite side and being unmatched (represented by matching to oneself). Formally, each $m \in M$ ranks $W \cup \{m\}$, and each $w \in W$ ranks $M \cup \{w\}$. Let $p \equiv (\succ_i)_{i \in M \cup W}$ denote the preference profile. For any agent $i$, let $P(i)$ denote the set of potential partners for $i$, consisting of the entire opposite side and $\{i\}$. For $j, k \in P(i)$, we write $j \succ_i k$ if agent $i$ strictly prefers being matched to $j$ rather than to $k$.

A matching is a mapping $\mu : M \cup W \to M \cup W$ such that: (i) $\mu(m) \in W \cup \{m\}$ for all $m \in M$; (ii) $\mu(w) \in M \cup \{w\}$ for all $w \in W$; and (iii) $\mu(i) = j$ if and only if $\mu(j) = i$ for all $i, j \in M \cup W$. We write $\mu(i) = i$ to indicate that agent $i$ is unmatched under $\mu$.[1]

For two matchings $\mu, \mu'$, write $\mu \succeq_W \mu'$ if every $w \in W$ weakly prefers $\mu$ to $\mu'$, that is, $\mu(w) \succeq_w \mu'(w)$ for all $w \in W$. Define $\mu \succeq_M \mu'$ analogously.

**Definition 1** (Blocking and stability)**.**
*A matching $\mu$ is* blocked *by an individual $i$ if $\mu(i) \neq i$ but $i \succ_i \mu(i)$.*
*A matching $\mu$ is* blocked *by a pair $(m, w) \in M \times W$ if $m \succ_w \mu(w)$ and $w \succ_m \mu(m)$.*
*A matching $\mu$ is* stable *if it is not blocked by any individual or pair.*

These are definitions from Gale and Shapley [1962] for static markets.

[1] Throughout the paper, when we say an agent is matched, we always mean matched to an agent on the opposite side of the market.

In this paper, we allow the possibility of future entry or exit of agents which will transform our current market $(M, W, p)$ to a new perturbed market $(M', W', p')$, where $p'$ is the preference profile of agents in the perturbed market.

Let $\mathcal{S}(M, W, p)$ denote the set of stable matchings of market $(M, W, p)$.

**Definition 2** (Pair-preserving stability)**.** *A stable matching $s \in \mathcal{S}(M, W, p)$ is* pair-preserving stable *against a perturbed market $(M', W', p')$ if there exists a stable matching $s' \in \mathcal{S}(M', W', p')$ such that for every $(m, w) \in M \times W$,*

$$s(m) = w \text{ and } (m, w) \in M' \times W' \quad \Longrightarrow \quad s'(m) = w.$$

That is, every pair matched under $s$ whose members both remain in the market must remain matched under $s'$.

We impose a standing assumption on preferences in the perturbed market. Let $p' \equiv (\succ'_i)_{i \in M' \cup W'}$ denote the preference profile in the perturbed market.

Throughout the paper, we assume that $p'$ agrees with $p$ on all agents who appear in both markets. Formally, for any $i \in (M \cap M') \cup (W \cap W')$ and any $j, k \in P(i)$ who also appear in both markets,

$$j \succ_i k \quad \Longrightarrow \quad j \succ'_i k.$$

Thus, relative rankings among surviving agents are unchanged. Let $\mathcal{P}'$ denote the set of preference profiles on $(M', W')$ that satisfy this condition.

Under this assumption, uncertainty arises only from: (i) how agents rank new arrivals relative to existing agents; (ii) the preferences of newly arriving agents; and (iii) which agents exit the market.

# 3 Entry on One Side of the Market

This section studies pair-preserving stability under one-sided entry. We characterize the blocking structure of pair-preserving extensions and identify the stable matching that is maximally pair-preserving stable. We then derive a tractable criterion for existence based on a single critical extension.

## 3.1 Stability Analysis Under One-Sided Entry

Without loss of generality, suppose that only men enter the market, so that $M' \supseteq M$ and $W' = W$. The perturbed preference profile $p'$ agrees with $p$ on all original agents. Uncertainty therefore concerns only preferences involving newly arriving men.

Under one-sided entry, every pair matched in a stable matching of the original market remains feasible in the expanded market. Consequently, pair-preserving stability depends solely on whether originally unmatched women can be matched to newly arriving men in a way that yields stability in the expanded market.

For any stable matching $s_i \in \mathcal{S}(M, W, p)$, let $E(s_i) \subseteq M$ and $\hat{E}(s_i) \subseteq W$ denote the sets of men and women matched under $s_i$.

In a one-to-one matching market, the Rural Hospitals Theorem [Roth, 1986] establishes that $E(s_i) = E(s_j)$ and $\hat{E}(s_i) = \hat{E}(s_j)$ for all $s_j, s_i \in \mathcal{S}(M, W, p)$. Thus, the set of matched agents is identical across all stable matchings, and we can simplify our notation to $E(s)$ and $\hat{E}(s)$ without ambiguity.

To organize the analysis, we decompose the expanded market (Figure 1) and characterize where blocking pairs may arise.

The unshaded regions represent agents matched in the original market, whose matches must be preserved. The shaded regions form a free submarket in which new matches may be selected to achieve stability in the expanded market.

Any matching $h$ on $(M', W, p')$ that preserves a stable matching $s \in \mathcal{S}(M, W, p)$ and satisfies individual rationality (IR) must take the following form:

$$h(i) = \begin{cases} s(i), & \text{if } i \in E(s) \cup \hat{E}(s), \\ g(i), & \text{if } i \in (M' \setminus E(s)) \cup (W \setminus \hat{E}(s)), \end{cases}$$

where $g$ is a matching on the free submarket $(M' \setminus E(s)) \cup (W \setminus \hat{E}(s))$. We require $g(i) \succeq_i i$ for all agents in the free submarket so that $h$ satisfies IR.

Although the free submarket set $(M' \setminus E(s)) \cup (W \setminus \hat{E}(s))$ includes originally unmatched men, stability of $s$ implies that no such man can form a mutually acceptable match with any woman in $W \setminus \hat{E}(s)$. Consequently, in any IR extension preserving $s$, agents in $M \setminus E(s)$ must remain unmatched.

**Lemma 1.** *Let $s$ be a stable matching in $(M, W, p)$, and let $h$ be any matching in $(M', W, p')$ that preserves all matched pairs in $s$ and satisfies IR. Then any blocking*

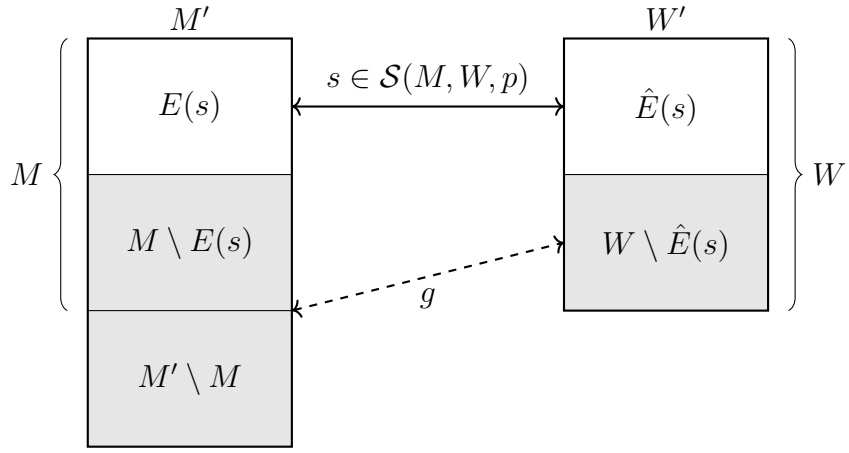


Figure 1: Decomposition of the expanded market into originally matched agents and the free submarket (one-sided entry).

*pair of $h$ (if any exist) must involve a newly arriving man $m \in M' \setminus M$ and a woman (matched or single).*

The full proof appears in Appendix A.1. The argument is straightforward. Since $s$ is stable in the original market $(M, W, p)$ and the preferences among the original agents are unchanged in $p'$, no pair of original agents can block $h$ in the expanded market. Hence, any blocking pair must involve at least one new agent. Because only men enter, any such blocking pair must consist of a new man and a woman who may be either previously matched or previously unmatched.

**Theorem 1.** *Let $s_W \in \mathcal{S}(M, W, p)$ denote the woman-optimal stable matching. If $p' \in \mathcal{P}'$ is a perturbed preference profile such that $s_W$ is not pair-preserving stable against $(M', W, p')$, then no stable matching $s \in \mathcal{S}(M, W, p)$ is pair-preserving stable against $(M', W, p')$.*

The full proof is in Appendix A.2; we sketch the main argument. If $s_W$ is not pair-preserving stable against $(M', W, p')$, then no stable matching of the expanded market preserves all pairs of $s_W$. Equivalently, every extension of $s_W$ to $(M', W, p')$ that satisfies IR admits a blocking pair. For convenience, call any such matchings an IR extension of $s_W$.

By the Rural Hospitals Theorem, all stable matchings of the original market induce the same partition of agents into matched and unmatched sets. Hence, for fixed $p'$, the free submarket is identical across all stable matchings $s \in \mathcal{S}(M, W, p)$. Thus, the

set of feasible IR matchings on the free submarket is the same for all $s$. Accordingly, every IR extension of any stable matching $s$ can be written as the original matching $s$ combined with an IR matching on this common free submarket.

By Lemma 1, any blocking pair in an IR extension of $s_W$ must involve a new man and a woman. If a blocking pair consists of a new man and a woman matched under $s_W$, then the same pair blocks the corresponding IR extension of any other stable matching $s$. Since $s_W$ is woman-optimal, every woman weakly prefers her partner in $s_W$ to her partner in any other stable matching [Gale and Shapley, 1962]. Thus, if she prefers the new man to her partner in $s_W$, she also prefers him to her partner in $s$. The new man's incentive to deviate is determined by $p'$ only and is not affected by the original stable matching.

If instead the blocking pair involves a new man and a previously unmatched woman, the same blocking pair will also block the corresponding IR extension formed from any other $s$.

Therefore, if every IR extension of $s_W$ is blocked, then every IR extension of any stable matching is blocked. The woman-optimal stable matching is thus extremal with respect to pair-preserving stability under one-sided entry.

Theorem 1 identifies the woman-optimal stable matching as extremal with respect to pair-preserving stability under one-sided entry. By similar logic, man-optimal stable matching $s_M \in \mathcal{S}(M, W, p)$ is the least pair-preserving stable in the sense that if any stable matching $s \in \mathcal{S}(M, W, p)$ is not pair-preserving stable against a perturbation, $s_M$ will fail pair-preserving stability.

**Corollary 1.** *(Least Pair-Preserving Stable Matching) Let $s_M \in \mathcal{S}(M, W, p)$ denote the man-optimal stable matching. If $p' \in \mathcal{P}'$ is a perturbed preference profile such that there exists a $s \in \mathcal{S}(M, W, p)$ that is not pair-preserving stable against $(M', W, p')$, then $s_M \in \mathcal{S}(M, W, p)$ is not pair-preserving stable against $(M', W, p')$.*

The preceding argument extends naturally to settings in which agents may exit from the opposite side of the market.

**Corollary 2** (Entry and Exit)**.** *Suppose that the market changes from $(M, W)$ to $(M', W')$ with $M \subseteq M'$ and $W \supseteq W'$, so that only men enter and only women exit. Suppose further that every woman who leaves, $w \in W \setminus W'$, has the same partner across all stable matchings of $(M, W, p)$. If the woman-optimal stable matching $s_W$ is*

*not pair-preserving stable against* $(M', W', p')$*, then no stable matching in* $\mathcal{S}(M, W, p)$ *is pair-preserving stable against* $(M', W', p')$*.*

This result extends Theorem 1 to allow the departure of women with invariant partners.

(i) If she was single (in which case, her invariant partner is herself), her departure may alter the matching within the free submarket. However, because she is single in every stable matching of $(M, W, p)$, the induced change in the free submarket is identical across all initial stable matchings. Hence the comparison argument used in Theorem 1 continues to apply.

(ii) If she exits while matched, her partner becomes unmatched. Under the invariant-partner assumption, the same man becomes free across all stable matchings of the original market. This again leaves the induced free submarket identical across all initial stable matchings, allowing the comparison argument of Theorem 1 to carry through.

By contrast, if women with variant partners were allowed to exit, different stable matchings would free different men, and the comparison argument would fail. Hence, no analogous uniform conclusion can be drawn in that case.

The previous results compare pair-preserving stability across stable matchings within a fixed market. We now consider how pair-preserving stability varies across nested markets.

**Corollary 3** (Nested Markets)**.** *Consider two markets* $(M, W, p)$ *and* $(\tilde{M}, W, \tilde{p})$ *with* $M \subseteq \tilde{M}$*, where* $\tilde{p}$ *agrees with* $p$ *on all common agents. Suppose* $s \in \mathcal{S}(M, W, p)$ *and* $\tilde{s} \in \mathcal{S}(\tilde{M}, W, \tilde{p})$ *match the same set of women, so that* $\hat{E}(\tilde{s}) = \hat{E}(s)$*. Assume the same future entry occurs in both markets:* $\tilde{M}' \setminus \tilde{M} = M' \setminus M$*, with* $\tilde{p}'$ *agreeing with* $p'$*. If no pair-preserving stable matching exists for* $(\tilde{M}, W, \tilde{p})$ *against* $(\tilde{M}', W, \tilde{p}')$*, then none exists for* $(M, W, p)$ *against* $(M', W, p')$*.*

When both markets select woman-optimal stable matchings, every woman is weakly better off in the larger market [Crawford, 1991]. Intuitively, a larger initial pool of men gives each woman a partner who is at least as desirable as her partner in the smaller market. Holding the free submarkets fixed, any blocking pair for an IR extension of the woman-optimal stable matching in the larger market also blocks the corresponding extension in the smaller market.

## 3.2 Application of Results

Having established that the woman-optimal stable matching $s_W$ is maximally pair-preserving stable to one-sided entry of men, we now consider the problem faced by a planner who anticipates such entry. When the planner expects additional men to arrive but is uncertain about the realized preference profile $p'$, she must choose a stable matching in the current market that maximizes the likelihood of being extendable to a stable matching after entry.

**Problem 1.** *Consider the following two-period central planner problem.*

*At $t = 0$, the market is $(M, W, p)$. The planner correctly anticipates that at $t = 1$ only men will enter, so that the market becomes $(M', W, p')$. However, she has no information about the set of entrants $M' \setminus M$ or the realized preferences $p'$ beyond the assumption that $p' \in \mathcal{P}'$.*

*The planner must choose:*

- *a stable matching $s_i \in \mathcal{S}(M, W, p)$ at $t = 0$, and*
- *a stable matching $s'_i \in \mathcal{S}(M', W, p')$ at $t = 1$, subject to the constraint that $s'_i$ preserves all matched pairs in $s_i$.*

Although such a pair $(s_i, s'_i)$ may not exist for all realizations of $(M', W, p')$, the planner may nevertheless wish to select a matching at $t = 0$ that maximizes the likelihood of extendability. Motivated by Theorem 1, we now present an algorithm that constructs the matching that is closest to stability in the expanded market while preserving all previously formed pairs.

**Algorithm 1.** ***Step 1 (t = 0).*** *Run the woman-proposing deferred acceptance algorithm on $(M, W)$ to obtain the woman-optimal stable matching $s_W$ of the initial market.*

***Step 2 (t = 1).*** *After new men arrive, run the man-proposing deferred acceptance algorithm on the submarket $(W \setminus \hat{E}(s), M' \setminus M)$ consisting of originally unmatched women and newly arriving men. This yields the man-optimal stable matching $\hat{s}_M$ of the submarket.*

***Step 3.*** *Combine $s_W$ and $\hat{s}_M$ to form a matching on the full market $(M', W)$. Denote this combined matching by $s_W \cup \hat{s}_M$.*

Let $\mathcal{S}_0 := \mathcal{S}(M, W, p)$ denote the set of stable matchings of the original market. Let $\hat{\mathcal{S}} := \mathcal{S}(M' \setminus M, W \setminus \hat{E}(s), \hat{p})$ denote the set of stable matchings of the submarket at $t = 1$, where $\hat{p}$ agrees with $p'$ and only concerns agents in the submarket.

For any $s \in \mathcal{S}_0$ and any $\hat{s} \in \hat{\mathcal{S}}$, the matching $s \cup \hat{s} \in \mathcal{S}_0 \times \hat{\mathcal{S}}$ is a well-defined matching of the expanded market $(M', W, p')$. The two components do not conflict: $s$ specifies matches among originally matched agents, while $\hat{s}$ specifies matches within the submarket. Consequently, $\mathcal{S}_0 \times \hat{\mathcal{S}}$ is the set of matchings of the expanded market that (i) preserve all pairs of some stable matching in the original market and (ii) are stable within the submarket of originally single women and newly arriving men.

**Lemma 2.** *For every $p' \in \mathcal{P}'$, the matching $s_W \cup \hat{s}_M$ produced by Algorithm 1 minimizes the number of blocking pairs among all matchings in $\mathcal{S}_0 \times \hat{\mathcal{S}}$.*

The formal proof appears in Appendix A.3; here we provide the intuition. Restricting attention to $\mathcal{S}_0 \times \hat{\mathcal{S}}$ ensures that all originally single women and all newly arriving men are matched according to a stable matching of the submarket, thus eliminating blocking pairs between these agents. Any remaining instability must therefore arise from blocking pairs between a new man and a woman matched in the original market.

Pair-preserving stability thus depends on two forces: (i) the incentive of a matched woman to deviate with a new man, and (ii) the incentive of a new man to deviate with a matched woman rather than remain with his submarket partner. The matching $s_W \cup \hat{s}_M$ minimizes both. The woman-optimal matching $s_W$ makes each matched woman as well off as possible, and the man-optimal submarket matching $\hat{s}_M$ makes each new man as well off as possible. Consequently, $s_W \cup \hat{s}_M$ minimizes the number of blocking pairs among all matchings in $\mathcal{S}_0 \times \hat{\mathcal{S}}$.

This yields an immediate and useful implication.

**Proposition 1** (Existence of Pair-Preserving Stable Matchings)**.** *There exists a pair-preserving stable matching $s \in \mathcal{S}(M, W, p)$ against the perturbation $(M', W, p')$ if and only if the combined matching $s_W \cup \hat{s}_M$ is stable in the expanded market; that is,*

$$\exists s \text{ that is pair-preserving stable} \iff s_W \cup \hat{s}_M \in \mathcal{S}(M', W, p').$$

Lemma 2 implies that pair-preserving stability needs only be evaluated within $\mathcal{S}_0 \times \hat{\mathcal{S}}$, and that $s_W \cup \hat{s}_M$ is the element of this set with the fewest blocking pairs.

If this matching fails to be stable in the expanded market, then no matching is pair-preserving stable. Conversely, if $s_W \cup \hat{s}_M$ is stable, $s_W$ is pair-preserving stable.

Therefore, Proposition 1 shows that the existence of a pair-preserving stable matching can be determined by examining a single matching in the expanded market. This critical matching, $s_W \cup \hat{s}_M$, provides a tractable test of feasibility.

# 4 Entry on Both Sides of the Market

When both sides of the market expand, pair-preserving stability becomes more complex than in the one-sided entry case. Newly arriving agents on both sides may form blocking pairs with agents matched in the original market, so a planner faces competing considerations. Making women better off at $t = 0$ reduces the likelihood that matched women block with new men, but simultaneously makes matched men worse off and increases the likelihood that they block with new women. Consequently, unlike in one-sided entry, no stable matching is uniformly most pair-preserving stable under all perturbations.

To obtain a general result, we impose an additional assumption on the preference structure. The main result for the two-sided entry case is summarized in Theorem 2. The remainder of the section introduces the market decomposition, characterizes the relevant blocking pairs, and develops a probabilistic framework for comparing stability.

Before stating the main result, recall that for any stable matching $s$ of the original market and any matching $g$ of the submarket of free agents at $t = 1$, we write $s \cup g$ for the combined matching that preserves all pairs of $s$ and applies $g$ on free agents.

We assume—as formalized later in this section— valuations concerning newly arriving partners are drawn i.i.d. from a uniform distribution. Preferences are induced by these valuations.

For an arbitrary matching $h$ of the expanded market $(M', W', p')$, define the event

$$A(h) \equiv \{h \text{ is stable under } (M', W', p')\}.$$

Throughout this section, probabilities are taken conditional on the realized preferences that generate the stable matching $s_i$ in the original market and the submarket matching $g$. The remaining uncertainty therefore arises only from valuations between

agents in the original market and agents in the submarket.

**Theorem 2** (Asymmetric entry)**.** *Consider a sequence of expanded markets indexed by $k$ in which $\Delta n_k$ new women and $\Delta n_k + d_k$ new men enter at $t = 1$, with $d_k/\Delta n_k \to \infty$. Assume valuations involving newly arriving agents are drawn i.i.d. from $U[0,1]$. As $k \to \infty$, for any matching $g$ in the submarket, the woman-optimal stable matching $s_W$ maximizes the probability that the combined matching*

$$h_W \;=\; s_W \cup g$$

*is stable in the merged market $(M', W', p')$. That is, $\forall s_i \in \mathcal{S}(M, W, p)$ and $\forall$ submarket matching $g$,*

$$\mathbb{P}[A(h_W = s_W \cup g)] \geq \; \mathbb{P}[A(h_i = s_i \cup g)].$$

Choosing $s_W$ at $t = 0$ makes each originally matched woman as well off as possible among all stable matchings, raising her threshold to block with new men. The same choice makes originally matched men weakly worse off, increasing their propensity to block with new women. Given this innate trade-off, the key insight in Theorem 2 is that, under highly asymmetric entry, blocking by new men is the dominant threat to stability. Hence, when substantially more men than women enter, the planner should prioritize mitigating new men blocking pairs by selecting the woman-optimal stable matching at $t = 0$. The proof is in Appendix A.6.

**Remarks.** (i) The inequality in Theorem 2 holds for every realization of the submarket (subject to the entry-size condition) and for every stable matching of the free submarket, $g \in \hat{\mathcal{S}}$. (It holds at equality when $g$ is not stable.) (ii) Theorem 2 extends the logic of Theorem 1. If only men enter, Theorem 1 implies that choosing $s_W$ at $t = 0$ maximizes the prospects of obtaining a stable extension at $t = 1$. Theorem 2 shows that the same guidance applies under two-sided entry when the inflow of men is sufficiently larger than the inflow of women.

We now derive the analysis for two-sided entry. Starting from $(M, W, p)$, the market expands to $(M', W', p')$ with

$$M \subseteq M', \qquad W \subseteq W',$$

where the perturbed profile $p' \equiv (\succ_i')_{i \in M' \cup W'}$ agrees with $p$ on all agents who appear in both markets.

Figure 2 decomposes the expanded market.

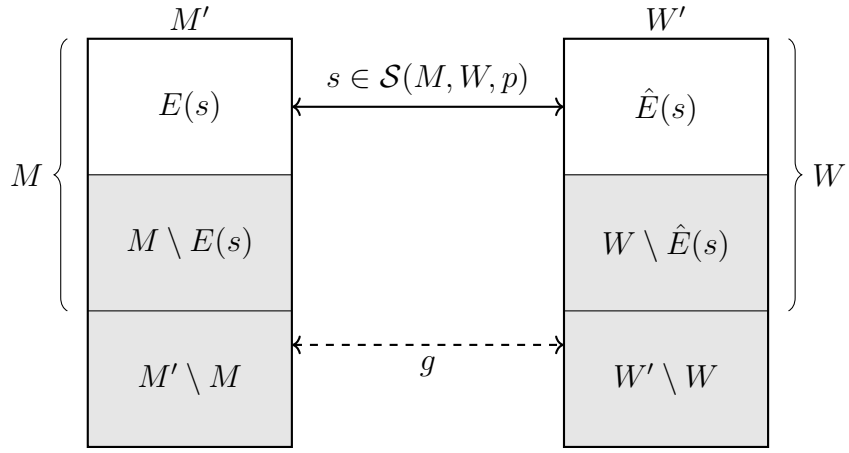


Figure 2: Decomposition of the expanded market into originally matched agents and the free submarket (two-sided entry).

Fix a stable matching $s \in \mathcal{S}(M, W, p)$ in the initial market. Let $E(s) \subseteq M$ and $\hat{E}(s) \subseteq W$ denote the sets of men and women matched under $s$. When the market expands to $(M', W', p')$, agents fall into three groups:

- originally matched agents $E(s)$ and $\hat{E}(s)$;
- originally single agents $M \setminus E(s)$ and $W \setminus \hat{E}(s)$;
- newly arriving agents $M' \setminus M$ and $W' \setminus W$.

The free submarket, composed of originally single and newly arriving agents, is shaded in Figure 2.

A matching $h$ of the expanded market that *preserves* a stable matching $s \in \mathcal{S}(M, W, p)$ has the following structure:

$$h(i) = \begin{cases} s(i), & \text{if } i \in E(s) \cup \hat{E}(s), \\ g(i), & \text{if } i \in (M' \setminus E(s)) \cup (W' \setminus \hat{E}(s)), \end{cases}$$

Any blocking pair of $h$ must involve at least one agent from the submarket. And, to eliminate blocking pairs within the submarket, restrict $g$ to be a stable matching of the submarket. Any remaining blocking pair in the full market must be a cross pair between an originally matched agent and a free agent, as stated formally in Lemma 3.

**Algorithm 2.** ***Step 1 (t = 0).*** *Choose a stable matching* $s \in \mathcal{S}_0 := \mathcal{S}(M, W, p)$.
***Step 2 (t = 1).*** *Choose a stable matching* $g \in \hat{\mathcal{S}} := \mathcal{S}(W' \setminus \hat{E}(s),\ M' \setminus E(s),\ \hat{p})$.
***Step 3.*** *Combine* $s$ *and* $g$ *to form* $h = s \cup g$ *on* $(M', W', p')$.

**Lemma 3.** *Let* $s \in \mathcal{S}(M, W, p)$ *be a stable matching of the original market, and let* $h$ *be constructed as in Algorithm 2. Then any blocking pair* $(m, w)$ *of* $h$ *is of one of the following two types:*

**Type 1.** $w \in \hat{E}(s)$ *and* $m \in M' \setminus M$*: a previously matched woman and a new man.*

**Type 2.** $w \in W' \setminus W$ *and* $m \in E(s)$*: a new woman and a previously matched man.*

Timing. At $t = 0$, the planner observes the initial market $(M, W, p)$ and selects a stable matching $s \in \mathcal{S}(M, W, p)$, knowing only the distribution of future valuations. At $t = 1$, the preference profile within the free submarket, $\hat{p}$, is realized. Restricting attention to $\hat{\mathcal{S}}$ is without loss, since any extension outside this set induces instability in the merged market. The planner thus selects a stable submarket matching $g \in \hat{\mathcal{S}}$ based on the realized $\hat{p}$. Valuations across the original market and the submarket are realized only after the merger of the two markets.

Under one-sided entry, the comparison across stable matchings does not require assumptions on $p'$ (beyond that it agrees with $p$). In contrast, with two-sided entry, the side the planner should prioritize depends on $p'$. For example, suppose that all new men find only free women acceptable (either newly arriving or originally single), while all new women strictly prefer to match with matched men. Then, Type 1 blocking is impossible, and to reduce Type 2 blocking, the planner should choose man-optimal stable matching at $t = 0$. A symmetric construction reverses the conclusion. These polar cases motivate imposing structure on preferences to obtain general comparative result.

We introduce a valuation representation. Each agent assigns a valuation to every potential partner. For $i, j \in M \cup W$, let $v_i(j) \in [0, 1]$ denote $i$'s valuation for matching with $j$, and normalize the valuation of being single to zero: $v_i(i) = 0$. Preferences are induced by valuations via

$$v_i(j) > v_i(k) \quad \Longleftrightarrow \quad j \succ_i k.$$

Uncertainty about $p'$ arises from valuations involving newly arriving agents. Assume for each original agent $i \in M \cup W$ and each new agent $j \in (M' \setminus M) \cup (W' \setminus W)$,

the valuations $v_i(j)$ are drawn i.i.d. from $U[0,1]$. For each new agent $i \in (M' \setminus M) \cup (W' \setminus W)$ and each potential partner $j \in M' \cup W'$, the valuations $v_i(j)$ are drawn i.i.d. from $U[0,1]$.

For any agent $i$ and matching $s$, define the *threshold for blocking* as

$$b_i(s) \equiv v_i\big(s(i)\big),$$

with $b_i(s) = 0$ whenever $i$ is single under $s$. Then $i$ prefers $j$ to her partner under $s$ if and only if $v_i(j) > b_i(s)$.

We can now express the probability that a combined matching $h_i = s_i \cup g \in \mathcal{S}_0 \times \hat{\mathcal{S}}$ is stable in the merged market. This probability is conditional on $s_i$ and $g$.

**Lemma 4.** *Fix a stable matching $s_i \in \mathcal{S}(M, W, p)$ and a stable submarket matching $g \in \hat{\mathcal{S}}$. Conditional on the realized preferences that generate $s_i$ and $g$, the probability that $h_i = s_i \cup g$ is stable in the merged market $(M', W', p')$ is $\mathbb{P}\big[A(h_i = s_i \cup g)\big] =$*

$$\prod_{(w,m)\in\hat{E}(s_i)\times(M'\setminus M)} \big[1-(1-b_m(g))(1-b_w(s_i))\big] \prod_{(m,w)\in E(s_i)\times(W'\setminus W)} \big[1-(1-b_w(g))(1-b_m(s_i))\big]. \tag{3}$$

The proof is in Appendix A.5. Lemma 4 expresses, for any initial stable matching $s_i$ and any stable submarket matching $g$, the exact probability that $h_i = s_i \cup g$ avoids blocking. The comparison in Theorem 2 fixes an arbitrary $g$ and evaluates $\mathbb{P}[A(h_i = s_i \cup g)]$ across $s_i \in \mathcal{S}(M, W, p)$. In the asymmetric-entry regime, the woman-optimal matching $s_W$ maximizes this stability probability for every submarket matching $g$. Hence, the planner should select $s_W$, knowing that for any realized submarket and any realized $g$, the combined matching $s_W \cup g$ is most likely to be stable in the merged market.

Theorem 2 guides the planner's choice of the initial stable matching $s_i$ at $t = 0$. Once $s_i$ is fixed and the submarket is realized at $t = 1$, the planner should select $g$ to maximize the stability probability given in (3).

# 5 Conclusion

This paper introduces *pair-preserving stability*, a robustness criterion for stable matchings in markets subject to uncertain entry. The concept requires that previously

formed matches remain intact, and studies which stable matchings are most likely to admit stable extensions as the market expands.

Under one-sided entry of men (and one-sided exit of women with invariant partners), we obtain a sharp monotonicity result. The woman-optimal stable matching is maximally pair-preserving stable, while the man-optimal matching is the least. This dominance holds uniformly across all realizations of entrants' number and preferences and is independent of the size or preference profile of the initial market. Moreover, the existence of a pair-preserving stable matching can be determined by examining a single *critical* matching, formed by combining the woman-optimal stable matching of the original market with the man-optimal stable matching of the new submarket. This provides a tractable diagnostic tool for planners anticipating one-sided entry.

When both sides of the market expand, robustness becomes more complex because blocking pairs may arise in either direction. We show that no stable matching uniformly minimizes instability across all preference profiles. Nevertheless, under a natural i.i.d. valuation structure, the planner's problem becomes tractable. In highly asymmetric expansions—where substantially more men than women enter—the dominant threat to stability comes from new men blocking with already matched women. In this environment, the woman-optimal stable matching again maximizes the probability of reaching stability in the expanded market.

Together, the results clarify how the structure of market entry shapes the relative robustness of stable matchings. They also offer practical guidance: planners who anticipate one-sided or strongly asymmetric entry should favor the stable matching that makes the opposite side as well off as possible ex ante. The paper also proposes a procedure that planners can follow to construct (or approximate) a stable matching of the perturbed market.

# A Proofs

## A.1 Proof of Lemma 1

*Proof.* The argument proceeds by ruling out all other possible blocking pairs.

If there exist $w \in \hat{E}(s)$ and $m \in E(s)$ that form a blocking pair of $h$ under $(M', W, p')$, then they would also block $s$ under $(M, W, p)$. This contradicts the fact that $s$ is a stable matching of $(M, W, p)$.

Since $h$ preserves all matched pairs of $s$, it cannot form matches between any originally unmatched man $m \in M \setminus E(s)$ and a woman who is matched in $s$. Moreover, since $h$ satisfies individual rationality, it cannot form new matches between any originally unmatched men and originally unmatched women (by stability of $s$). There are no new women entering, so all agents in $M \setminus E(s)$ must remain unmatched in an IR extension $h$. Thus, $h$ forms new matches only between agents in $M' \setminus M$ and agents in $W \setminus \hat{E}(s)$.

If there exist $w \in \hat{E}(s)$ and $m \in M \setminus E(s)$ that block $h$ under $(M', W, p')$, then they would also block $s$ under $(M, W, p)$, again contradicting stability.

Finally, all $w \in W \setminus \hat{E}(s)$ are weakly better off under $h$ than under $s$. If there were a blocking pair consisting of some $w \in W \setminus \hat{E}(s)$ and some $m \in M$, the same pair would also block $s$, contradicting the stability of $s$. $\square$

## A.2 Proof of Theorem 1:

The proof relies on the following result of Gale and Shapley [1962]: $s_W \succeq_W s_j$ and $s_M \succeq_M s_j$ for all $s_j \in \mathcal{S}(M, W, p)$. That is, the woman-optimal stable matching is weakly preferred by all women to any other stable matching, and the man-optimal stable matching is weakly preferred by all men to any other stable matching.

*Proof.* Consider an arbitrary stable matching $s_j \in \mathcal{S}(M, W, p)$ that is not pair-preserving stable against $(M', W, p')$. By definition, this means that for every matching $h_j$ of $(M', W, p')$ that agrees with $s_j$ and satisfies individual rationality (IR), there exists a blocking pair. That is,

$$\exists w \in W,\ \exists m \in M' \text{ such that } m \succ_w h_j(w) \text{ and } w \succ_m h_j(m). \tag{1}$$

Any such matching $h_j$ has the following structure:

$$h_j(x) = \begin{cases} s_j(x), & x \in \hat{E}(s) \cup E(s), \\ g_j(x), & x \in (W \setminus \hat{E}(s)) \cup (M' \setminus E(s)), \end{cases}$$

where $g_j$ is a matching on the free submarket $(W \setminus \hat{E}(s)) \cup (M' \setminus E(s))$ satisfying $g_j(x) \succeq_x x$ for all agents in this set.

For a given $p'$ and stable matching $s_j$, let $H_j(p')$ denote the set of all such matchings $h_j$.

By the Rural Hospitals Theorem, the set $(W \setminus \hat{E}(s)) \cup (M' \setminus E(s))$ is invariant across all stable matchings $s_j \in \mathcal{S}(M, W, p)$. Hence, fixing $p'$, the set of feasible submarket matchings $g_j$ is the same for all stable matchings of the original market.

Consequently, for any $s_j, s_i \in \mathcal{S}(M, W, p)$ and any $h_j \in H_j(p')$, there exists a unique matching $h_i \in H_i(p')$ such that $h_j$ and $h_i$ coincide on all agents in $(W \setminus \hat{E}(s)) \cup (M' \setminus E(s))$ and differ only on $\hat{E}(s) \cup E(s)$.

Thus, for a fixed $p'$, there is a one-to-one correspondence between the sets of extension matchings $H_j(p')$ and $H_i(p')$ for any two stable matchings $s_j$ and $s_i$.

Now suppose that $p' \in \mathcal{P}'$ is a perturbed preference profile such that $s_W$ is not pair-preserving stable against $(M', W, p')$. By definition, for every matching $h_W \in H_W(p')$ that agrees with $s_W$ and satisfies IR, there exists a blocking pair.

By Lemma 1, blocking pairs can only take two forms:

$$\exists w \in \hat{E}(s), \exists m \in M' \setminus M \text{ such that } m \succ_w s(w),\ w \succ_m g(m), \tag{2}$$

$$\exists w \in W \setminus \hat{E}(s), \exists m \in M' \setminus M \text{ such that } m \succ_w g(w),\ w \succ_m g(m). \tag{3}$$

Consider an arbitrary $h_W \in H_W(p')$. Suppose that at least one blocking pair is of type (2). Then there exist $(w, m)$ such that

$$m \succ_w s_W(w), \qquad w \succ_m g_W(m).$$

Let $h_j \in H_j(p')$ denote the extension corresponding to $h_W$ under the one-to-one mapping described above, so that $g_j = g_W$. Since $s_W \succeq_W s_j$ for all $s_j \in \mathcal{S}(M, W, p)$, we have

$$m \succ_w s_W(w) \ \Rightarrow\ m \succ_w s_j(w).$$

Together with $w \succ_m g_j(m)$, it follows that the same pair $(w, m)$ also blocks the corresponding matching $h_j$.

Now suppose instead that a blocking pair of $h_W$ is of type (3). Because the corresponding matching $h_j$ coincides with $h_W$ on the submarket matching $(g_j = g_W)$, the same pair $(w, m)$ also satisfies

$$m \succ_w g_j(w), \qquad w \succ_m g_j(m),$$

and therefore also blocks $h_j$.

Thus, for every $h_W \in H_W(p')$, the blocking pair that blocks $h_W$ also blocks the corresponding $h_j \in H_j(p')$. Since this holds for all possible extensions $h_W$, every extension $h_j$ admits at least one blocking pair.

Therefore, for all $s_j \in \mathcal{S}(M, W, p)$, every extension matching $h_j$ of $(M', W, p')$ that agrees with $s_j$ admits a blocking pair. Hence no stable matching $s_j \in \mathcal{S}(M, W, p)$ is pair-preserving stable against $(M', W, p')$. □

## A.3 Proof of Lemma 2

*Proof.* Any matching $\mu \in S \times \hat{S}$ agrees with a stable matching of $(M, W, p)$. By Lemma 1, any blocking pair of $\mu$ in $(M', W, p')$ must involve a man in $M' \setminus M$ and a woman in either $\hat{E}(s)$ or $W \setminus \hat{E}(s)$.

By construction, $\mu$ also agrees with a stable matching of the submarket $(W \setminus \hat{E}(s), M' \setminus M, \hat{p})$. This rules out blocking pairs involving women in $W \setminus \hat{E}(s)$. Hence, only pairs $(m, w)$ with $m \in M' \setminus M$ and $w \in \hat{E}(s)$ can block $\mu$.

Suppose $(m, w)$ blocks $s_W \cup \hat{s}_M$, that is, $m \succ_w s_W(w)$ and $w \succ_m \hat{s}_M(m)$. By [Gale and Shapley, 1962], the same pair blocks every $\mu \in S \times \hat{S}$. □

## A.4 Proof of Corollary 3

*Proof.* If $\tilde{s}_W$ is not pair-preserving stable against $(\tilde{M}', W, \tilde{p}')$, then by Proposition 1 there exists a pair $m \in \tilde{M}' \setminus \tilde{M}$ and $w \in W$ such that:

$$m \succ_w \tilde{s}_W(w) \qquad \text{and} \qquad w \succ_m \tilde{\hat{s}}_M(m),$$

where $\tilde{\hat{s}}_M$ is the man-optimal stable matching of the submarket consisting of new men and previously single women. By Crawford [1991], the woman-optimal stable matching is monotone with respect to expansions of the men's side: specifically,

$$\tilde{s}_W(w) \succeq_W s_W(w), \ \forall w \in W$$

meaning that all women weakly prefer their partner in the market with more men to their partner in a nested market with less men, given that both markets follow woman-optimal stable matching. Therefore,

$$m \succ_w \tilde{s}_W(w) \quad \Longrightarrow \quad m \succ_w s_W(w).$$

By assumption, the submarket is identical in the two cases, $\tilde{\hat{s}}_M = \hat{s}_M$. Thus, we conclude that $(m, w)$ also blocks the $\hat{s}_M$ extension of $s_W$. Hence, $s_W$ is not pair-preserving stable against $(M', W, p')$ and a pair-preserving stable matching does not exist. □

## A.5 Proof of Lemma 4

*Proof.* $\mathbb{P}[A(h_i) \mid s_i, g] = \mathbb{P}[\nexists$ blocking pairs of *Type (1∨2)*$\mid s_i, g]$. *Type 1* blocking pairs form between $w \in \hat{E}(s)$ and $m \in M' \backslash M$, $\forall$ such pair $(w, m)$, the probability that they do not block $= [1 - (1 - b_m(g))(1 - b_w(s_i))]$. That is, 1− probability of both of them having valuation of each other above their current partner. By iid assumption,

$$\prod_{(w,m) \in \hat{E}(s) \times (M' \backslash M)} [1 - (1 - b_m(g))(1 - b_w(s_i))]$$

accounts for the probability of having no *Type 1* blocking pairs. Similarly,

$$\prod_{(m,w) \in E(s) \times (W' \backslash W)} [1 - (1 - b_w(g))(1 - b_m(s_i))]$$

accounts for the probability of having no *Type 2* blocking pairs. Agents involved in *Type 1* do not overlap with agents involved in *Type 2*, by independence of preference between agents, we can take the product to indicate the probability of having no blocking pair of any *Type*. □

## A.6 Proof of Theorem 2

*Proof.* We want to show that $s_W$ maximizes the equation in Lemma 4. For simplicity, let $a_i(j) \equiv 1 - b_i(j)$.

That is, we want to show $\forall s$:

$$\sum_{(w,m) \in \hat{E}(s) \times (M' \backslash M)} ln[1 - (a_m(g))(a_w(s_W))] + \sum_{(m,w) \in E(s) \times (W' \backslash W)} ln[1 - a_w(g))(a_m(s_W))] \tag{1}$$

$$\geq$$

$$\sum_{(w,m)\in\hat{E}(s)\times(M'\backslash M)} ln[1-(a_m(g))(a_w(s))] + \sum_{(m,w)\in E(s)\times(W'\backslash W)} ln[1-a_w(g))a_m(s))] \quad (2)$$

Denote $\Delta E(g)$ as the set of newly added men that are matched under a stable submarket matching $g$, and $\Delta\hat{E}(g)$ as the set of newly added women that are matched under a stable submarket matching $g$.
Separate the *matched* and *unmatched* among the newly added agents:

$$\begin{aligned}
(1) = & \sum_{(w,m)\in\hat{E}(s)\times\Delta E(g)} ln[1-(a_m(g))(a_w(s_W))] + \sum_{(w,m)\in\hat{E}(s)\times[(M'\backslash M)\backslash\Delta E(g)]} ln[1-a_w(s_W)] \\
+ & \sum_{(m,w)\in E(s)\times\Delta\hat{E}(g)} ln[1-a_w(g))(a_m(s_W))] + \sum_{(m,w)\in E(s)\times[(W'\backslash W)\backslash\Delta\hat{E}(g)]} ln[1-a_m(s_W)] \\
= & \sum_{(w,m)\in\hat{E}(s)\times\Delta E(g)} ln[1-(a_m(g))(a_w(s_W))] + |(M'\backslash M)\backslash\Delta E(g)| \sum_{w\in\hat{E}(s)} ln[1-a_w(s_W)] \\
+ & \sum_{(m,w)\in E(s)\times\Delta\hat{E}(g)} ln[1-a_w(g))(a_m(s_W))] + |(W'\backslash W)\backslash\Delta\hat{E}(g)| \sum_{m\in E(s)} ln[1-a_m(s_W)],
\end{aligned}$$

Take the difference and show that the difference is $\geq 0$:

$$\begin{aligned}
(1)-(2) = & \sum_{(w,m)\in\hat{E}(s)\times\Delta E(g)} ln[1-(a_m(g))(a_w(s_W))] - ln[1-(a_m(g))(a_w(s))] \\
& + |(M'\backslash M)\backslash\Delta E(g)| \sum_{w\in\hat{E}(s)} ln[1-a_w(s_W)] - ln[1-a_w(s)] \\
& + \sum_{(m,w)\in E(s)\times\Delta\hat{E}(g)} ln[1-(a_w(g))(a_m(s_W))] - ln[1-(a_w(g))(a_m(s))] \\
& + |(W'\backslash W)\backslash\Delta\hat{E}(g)| \sum_{m\in E(s)} ln[1-a_m(s_W)] - ln[1-a_m(s)]
\end{aligned}$$

By Gale and Shapley [1962], the first two terms of (1)−(2) are nonnegative and the last two terms are nonpositive.

Because the initial market $(M, W, p)$ is fixed and finite, and preferences are strict, every originally matched agent under any stable matching receives a partner strictly preferred to remaining single. Hence, for every originally matched agent $i$ and every

stable matching $s \in \mathcal{S}(M, W, p)$, we have $b_i(s) > 0$ and thus $a_i(s) = 1 - b_i(s) < 1$.

Since the set of originally matched agents and the set of stable matchings are finite, there exist constants $\bar{a}_M, \bar{a}_W < 1$ such that $a_m(s) \leq \bar{a}_M$ for all $m \in E(s)$ and $a_w(s) \leq \bar{a}_W$ for all $w \in \hat{E}(s)$, uniformly over $s \in \mathcal{S}(M, W, p)$. Therefore, all logarithmic summands in (1)-(2) are uniformly bounded.

Now fix any $s \neq s_W$, and define

$$c(s) := \sum_{w \in \hat{E}(s)} \Big[ ln(1 - a_w(s_W)) - ln(1 - a_w(s)) \Big].$$

Since $s_W \succeq_W s$, each term in this sum is nonnegative. Moreover, because $s \neq s_W$, at least one woman strictly prefers $s_W$ to $s$, so $c(s) > 0$.

In the submarket, a newly added man can be matched only with a woman in $(W \setminus \hat{E}(s)) \cup (W' \setminus W)$. Hence,

$$|\Delta E(g)| \leq |W \setminus \hat{E}(s)| + \Delta n_k.$$

Since $|M' \setminus M| = \Delta n_k + d_k$, it follows that

$$|(M' \setminus M) \setminus \Delta E(g)| \geq (\Delta n_k + d_k) - (|W \setminus \hat{E}(s)| + \Delta n_k) = d_k - |W \setminus \hat{E}(s)|.$$

Thus, the second term is bounded below by

$$(d_k - |W \setminus \hat{E}(s)|)\, c(s),$$

where $|W \setminus \hat{E}(s)|$ is fixed and finite.

By contrast, since $|W' \setminus W| = \Delta n_k$, we have

$$|\Delta \hat{E}(g)| \leq \Delta n_k, \qquad |(W' \setminus W) \setminus \Delta \hat{E}(g)| \leq \Delta n_k.$$

Because each logarithmic summand is uniformly bounded, there exists a finite constant $K(s)$ such that the sum of the absolute values of the third and fourth terms is at most $\Delta n_k K(s)$.

Since the first term is nonnegative, it follows that

$$(1) - (2) \geq (d_k - |W \setminus \hat{E}(s)|)\, c(s) - \Delta n_k K(s).$$

Therefore, if $\Delta n_k \neq 0$, $d_k/\Delta n_k \to \infty$, then for sufficiently large $k$,

$$(1) - (2) \geq 0.$$

This proves that $s_W$ maximizes the stability probability. □

When $\Delta n_k = 0$, the problem reduces to the one-sided entry case.

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